

EDDM: A Novel ECG Denoising Method Using Dual-Path Diffusion Model

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Abstract—Background: electrocardiogram (ECG), as a low-cost and noninvasive measurement tool, plays a crucial role in detecting arrhythmias. However, the ECG measurement process is highly vulnerable to factors such as baseline wandering (BW), electrode motion (EM), and muscle artifacts (MAs), leading to significant degradation in signal quality. Consequently, exploring denoising methods to obtain clean ECGs has emerged as an essential and challenging task. **Methods:** We propose a novel denoising method for ECG signals called the ECG denoising diffusion model (EDDM). Unlike traditional generative methods, EDDM employs two distinct pathways in the forward stage: ECG noise diffusion and Gaussian white noise diffusion. This unique approach allows for directed diffusion from the clean ECG domain to the measured ECG domain. In the reverse process, we design a U-Net model that combines deep aggregation pyramid pooling (DAPP) layers to accurately capture the multiscale information of the ECG for predicting the two types of noise during the diffusion process. Therefore, EDDM follows an interpretable denoising path and directly obtains the clean ECG from the noisy ECG instead of using a generative approach, resulting in more intuitive and stable denoising outcomes. **Conclusion:** EDDM has achieved state-of-the-art (SOTA) results on five metrics in benchmark experiments, with a significant improvement of 4%–10% in the percentage root-mean-square difference (PRD) compared to the existing SOTA model. Analysis shows that EDDM interpretably achieves the diffusion of clean ECG into noisy ECG, rather than white noise. Furthermore, under extreme noise conditions, EDDM achieves more stable and higher performance denoising results compared to the SOTA method, demonstrating the enormous potential of EDDM in wearable ECG measurement systems.

Index Terms—Convolutional neural network (CNN), diffusion probabilistic model, electrocardiogram (ECG) noise model, ECG, multiscale convolutional block, signal denoising.

I. INTRODUCTION

ELECTROCARDIOGRAM (ECG) serves as a noninvasive, portable, and cost-effective tool for clinical cardiac examinations, facilitating the capture of faint

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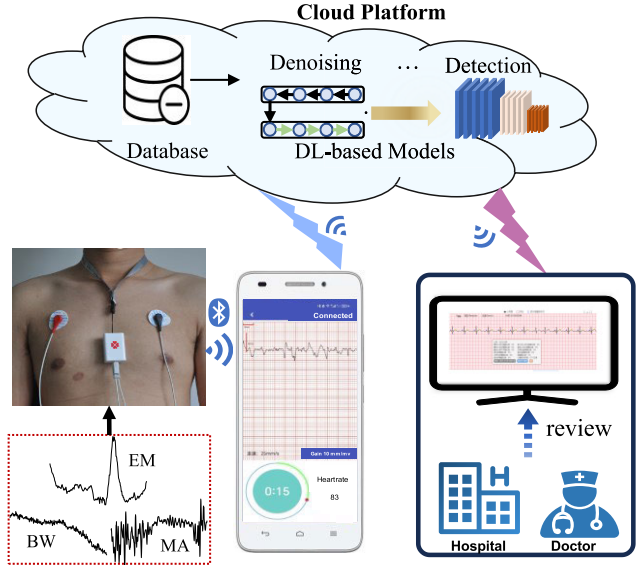


Fig. 1. Framework of an intelligent wearable ECG measurement system. (Noise types: BW, EM, and MA.)

electrical signals emitted by the heart [1], [2], [3]. This empowers cardiologists and clinical physicians with the opportunity to analyze the cardiac structure and variations in autonomous rhythms. Many studies [4], [5], [6] have shown that the widespread use of ECG examination is of great significance for improving the detection rate of arrhythmias. It enables the early discovery, early intervention, and early treatment of CVDs. In recent years, with the development of wearable devices and remote digital healthcare technology, many wearable intelligent ECG measurement systems [7], [8], [9] have been proposed and applied for out-of-hospital cardiac monitoring. As shown in Fig. 1, the wearer continuously measures their ECG signals and utilizes deep learning models on a cloud platform for real-time signal analysis. These systems allow doctors to remotely review the ECG and promptly identify any hidden abnormalities.

However, ECG signals are weak physiological electrical signals that are particularly vulnerable to interference from electrode movement and muscle tremors, especially in wearable devices. These noise signals can manifest as baseline wandering (BW), electrode motion (EM), and muscle artifacts (MAs) [10], [11], [12]. As seen in Fig. 1, BW noise appears as low-frequency fluctuations of 0.15–0.3 Hz in the ECG, and human respiration is one of its sources. EM noise is caused by

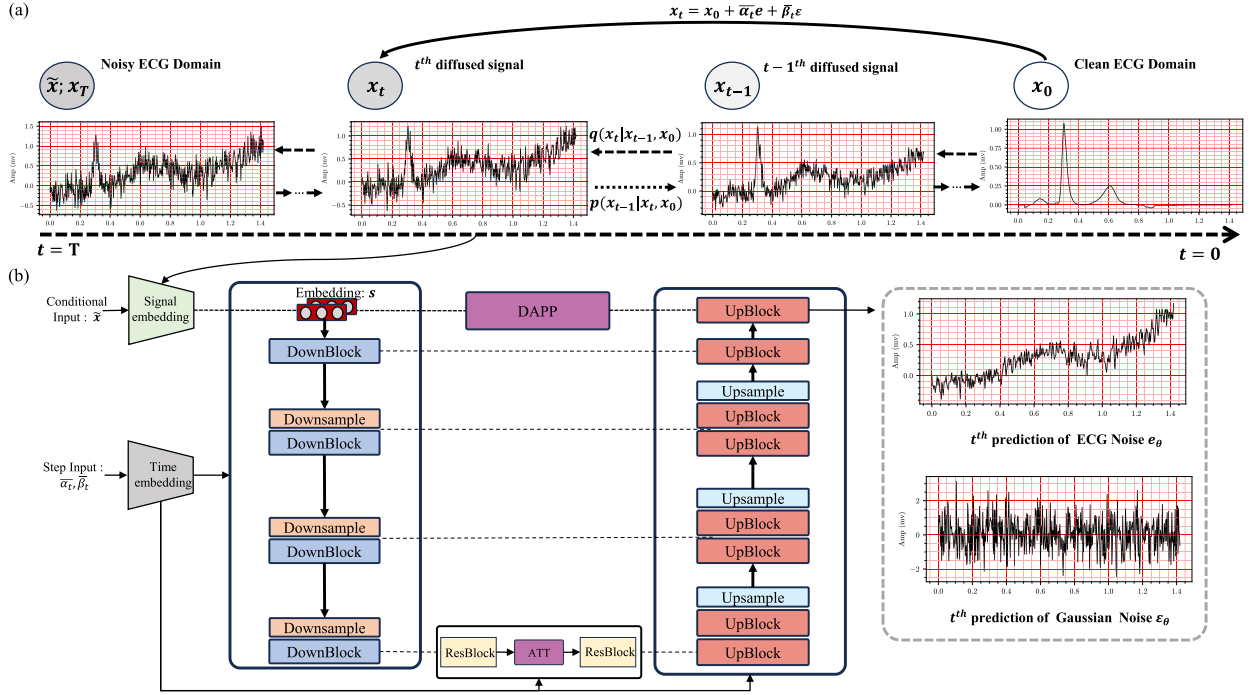


Fig. 2. Framework of EDDM. (a) Diffusion and reverse process of transforming clean ECG into noisy ECG in EDDM. (b) U-Net structure in EDDM to predict ECG noise e_θ and Gaussian noise ϵ_θ during the denoising (reverse) process at the t th step. (DAPP: DAPP layer. ATT: self-attention layer.)

the contact between electrodes and the skin, usually showing a random pattern with large fluctuations, leading to significant ECG changes. MA noise is generated by the skeletal muscle movements of the wearer, presenting diverse morphological changes with rapid trembling and jittering waveforms [13]. A more complex issue arises when these noises overlap with the frequency distribution of the ECG, becoming intermingled with ECG waveforms in the time domain [14]. In AI-based ECG systems, these noises reduce the accuracy of deep learning models, leading to higher rates of false negatives and positives [15]. Therefore, it is of great significance to develop ECG denoising algorithms and obtain high-quality signals, which is also a fundamental task in the field of AI-ECG [2].

Currently, ECG denoising algorithms consist of three main categories: traditional time-frequency denoising algorithms [16], [17], [18], [19], [20], deep learning methods based on denoising autoencoders [21], [22], [23], and generative denoising methods [24], [25]. Among them, a generative denoising model based on the denoising diffusion probabilistic model (DDPM) called DeScoD [25] has demonstrated state-of-the-art (SOTA) results compared to the other two types of methods. This method establishes a diffusion process from clean ECG to Gaussian white noise space, and in the reverse process, utilizes noisy ECG as conditions to generate “brand new” clean ECG, achieving “denoising.” However, there are still two issues that need to be addressed in the current generative denoising methods: 1) the instability of the generative denoising process, which produces uninterpretable outcomes, poses an inevitable limitation on the application within medical measurement systems. Particularly, for the DeScoD, the model unnecessarily introduces diffusion of the ECG into the Gaussian white noise space, thereby leading to

an uninterpretable denoising pathway for the ECG; and 2) there is room for improvement in the denoising performance based on the generative framework. It is worth exploring whether the sampling strategy from the white noise space is the optimal solution for the denoising problem.

Given the above two issues, this study explores a diffusion process more suitable for ECG denoising problems and proposes a more stable, interpretable, and higher performance ECG denoising diffusion model (EDDM). As shown in Fig. 2(a), we establish a novel diffusion process between the clean ECG and the measured ECG, instead of between clean ECG and Gaussian white noise space. In each step of this diffusion process, Gaussian white noise and real noise, two independent noise sources, are added to achieve the diffusion of the clean ECG along both a white noise diffusion path and an ECG noise diffusion path to the measured ECG space.

The primary contributions of this study are in three aspects.

- 1) Inspired by the inverse problem model for ECG denoising, we design a novel EDDM called EDDM. EDDM employs a unique dual-pathway diffusion process that simultaneously diffuses Gaussian white noise and ECG noise, achieving a directional path transformation between clean ECG and measured ECG. In comparison to existing generative denoising methods, the transition from noisy ECG signals to clean ECG signals is both stable and interpretable.
- 2) To mitigate the loss of ECG information and accumulation errors during the reverse prediction process, we design a 1-D residual U-Net that incorporates deep aggregation pyramid pooling (DAPP). This model extracts ECG features at multiple scales, capturing the

differences between the ECG signal and various types of noise.

- 3) The experimental results indicate that EDDM achieves SOTA performance across multiple metrics on three noise tasks. Compared with the existing SOTA model, EDDM shows an average improvement of 2.7% in maximum absolute distance (MAD) and a significant increase of 4%–10% in percentage root-mean-square difference (PRD). Cross-dataset experiments also demonstrate the excellent generalization ability of EDDM. Therefore, EDDM is expected to become the new benchmark model for ECG denoising.

II. RELATED WORKS

In this section, we elaborate on the current three major types of ECG denoising research and introduce the basic concept of ECG denoising modeling based on DDPM.

A. Research of ECG Denoising

Traditional ECG denoising research mainly focuses on designing filters to suppress noise frequency components and achieve denoising effects [17], [26]. These algorithms often target noise under specific conditions, and their effectiveness in removing high-noise and complex noise in real scenarios is not satisfactory.

In recent years, deep learning has rapidly developed in fields such as computer vision [27], time series prediction [28], [29], [30], and fault diagnosis [29], [31]. Similarly, many ECG denoising methods based on DAE have been proposed. Chiang et al. [23] and Jin et al. [13] developed an ECG DAE model using convolutional neural networks (CNNs). Antczak [32] and Kumar and Muduli [10] applied the deep recurrent neural network (DRNN) to accomplish end-to-end ECG denoising. Moreover, Romero et al. [33] enhanced the convolutional layer units in CNNs and introduced a deep learning model named DeepFilter. Overall, DAE-based deep learning denoising models achieve somewhat acceptable denoising results. In recent years, generative methods have been employed to investigate more efficient denoising techniques. Wang et al. [24] introduced a denoising approach for ECG based on conditional generative adversarial networks. This method involves training a discriminator and a generator alternately, enabling the generator, which is constructed using DAE, to generate clean ECG signals from noisy ones, thereby achieving superior denoising performance. Nevertheless, considerable potential exists for enhancement in denoising algorithms, and the models' interpretability remains inadequate.

B. Denoising Diffusion Probabilistic Model

DDPM is a generative model that has demonstrated excellent performance in tasks such as generation and denoising. DDPM consists of two processes: diffusion and reverse. In the diffusion process, the DDPM introduces Gaussian white noise to the input signal x_0 step by step, ultimately converting x_0 into the white noise space. Subsequently, through the

reverse process, the input signal is gradually recovered from the noise space. The concepts of diffusion and reversal are theoretically complementary, allowing the DDPM to acquire a connection between the input signal and the white noise space. Specifically, in the diffusion process, Gaussian noise is gradually added to the input signal x_0 , and each step of diffusion satisfies that the resulting signal follows a conditional Gaussian distribution. In other words, the signal x_t at the t th step satisfies the relationship with the signal x_{t-1} at the step $t - 1$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \gamma_t}x_{t-1}, \gamma_t \mathbf{I}) \quad (1)$$

where $\mathbf{I}$ is the identity matrix and γ_t represents hyperparameters, which increase with the number of diffusion steps t . They are used to compress the information of x_0 into x_t , so that the final x_T approximately follows a Gaussian distribution. Using the reparameterization strategy, the relationship between x_t and x_0 at any step in the diffusion process can be obtained

$$x_t = \sqrt{\bar{\xi}_t}x_0 + \sqrt{1 - \bar{\xi}_t}\varepsilon \quad (2)$$

where $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$, $\bar{\xi}_t = \prod_{i=1}^t \xi_i$, $\xi_i = 1 - \gamma_i$. During the forward process, there are no learning parameters involved. Instead, the diffusion for any step can be established by utilizing hyperparameters, Gaussian distribution, and the input signal x_0 .

In the reverse process, the derivation of [34] is obtained based on the assumption of a Markov chain

$$p_\theta(x_{t-1}|x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma(t)\mathbf{I}) \quad (3)$$

where $\mu_\theta(x_t, t)$ is characterized by learnable parameters, typically obtained by computing the noise ε_θ predicted from x_t, t using a deep learning network, while $\sigma(t)$ can be computed using hyperparameters γ_t . The training objective of DDPM is to minimize the difference between the predicted noise ε_θ at each step and the corresponding diffusion noise ε_t . The L_2 loss is commonly employed as the loss function for the optimization.

III. PROPOSED METHOD

A. Problem Definition

Considering the source of ECG noise [14], we assume that the acquired raw ECG signal consists of clean ECG, nonzero mean noise, and additive Gaussian white noise (AWGN)

$$\tilde{x} = x_0 + e + \eta \cdot \varepsilon \quad (4)$$

where x_0 represents the clean ECG signal, and e indicates the colored noise, such as BW, MA, and EW. $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$ represents the AWGN in the ECG signal $\tilde{x}$, and η represents the intensity of the ε . In the denoising task, we obtain the measured $\tilde{x}$ and desired to obtain $\hat{x} = x_0$. According to (4), there are many $(\hat{x}, e', n')$ satisfying the condition, which means x_0 and $\tilde{x}$ pose a many-to-one problem, or in other words, an inverse problem. When using deep learning for ECG denoising, we train a denoising model f_θ using noisy and clean ECG signals $\{\tilde{x}^k, x_0^k\}_{k=1}^N$. The optimization objective is to obtain $x_0 = f_\theta(\tilde{x})$. Thus, the denoising model f_θ should simultaneously model Gaussian white noise ε and nonzero mean noise e .

B. Dual-Path ECG Noise Diffusion Process

Considering (2) and (4), DDPM establishes a mapping relationship between white noise space ε and input signal x_0 in the diffusion process. However, x_0 does not autonomously diffuse the ECG-colored noise term e during the diffusion, resulting in the final x_T approximating a Gaussian distribution, without any specific association with the distribution of the noise ECG $\tilde{x}$.

Therefore, in Fig. 2(a), we use the dual-path ECG noise diffusion process to describe the transformation relationship from the original ECG x_0 to the noisy ECG $\tilde{x}$, in which both colored noise e and white noise ε are simultaneously added to x_0 in the diffusion process. This approach enables a comprehensible signal transformation path and, consequently, facilitates the construction of an ECG denoising model that aligns better with intuition. We define the diffusion process as follows:

$$\begin{aligned} x_t &= x_{t-1} + \alpha_t e + \beta_t \varepsilon_{t-1} \\ &= x_{t-2} + (\alpha_{t-1} + \alpha_t) e + \left(\sqrt{\beta_{t-1}^2 + \beta_t^2} \right) \varepsilon_{t-2} \\ &= \dots \\ &= x_0 + \bar{\alpha}_t e + \bar{\beta}_t \varepsilon \end{aligned} \quad (5)$$

where $e = \tilde{x} - x_0$ is defined as the ECG noise. In this study, we define $e \in \{\text{BW}, \text{EM}, \text{MA}\}$. $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon \sim \mathcal{N}(0, \mathbf{I})$ represent the Gaussian white noise added at each step during diffusion. α_t and β_t are hyperparameters used to control the diffusion strength of the colored noise e and the white noise ε , with $\bar{\alpha}_t = \sum_{i=1}^t \alpha_i$ and $\bar{\beta}_t = (\sum_{i=1}^t \beta_i^2)^{1/2}$. Therefore, based on (5), during the diffusion process, unlike DDPM, EDDM gradually adds Gaussian noise $\delta \sim \mathcal{N}(\alpha_t e, \beta^2 \mathbf{I})$ with nonzero mean to the original signal x_0 . This noise includes a certain proportion of colored noise e , such as BW, and a certain intensity of Gaussian white noise ε . This ensures that the signal does not spread into the Gaussian white noise space, but rather expands into a specific noise space. In particular, when $\bar{\alpha}_T = 1$, we can guarantee that the initial clean ECG x_0 finally diffuse to the noisy ECG signal, namely $x_T = \tilde{x} + \varepsilon$. Therefore, based on (5), we can establish an understandable diffusion process from clean ECG to noisy ECG.

In the reverse process, we need to recover x_0 from x_T . Since the diffusion process adds e and ε_{t-1} to x_{t-1} to obtain x_t , we introduce two models to predict the process from x_t to x_{t-1} with $e_\theta, \varepsilon_\theta$. Refer to [35] and [36], we derive the relationship between x_{t-1} and x_t . The detailed derivation process can be found in Appendix A. The probability distribution of both variables satisfies

$$\begin{aligned} p_\sigma(x_{t-1}|x_t, x_0, e) &= \mathcal{N}(x_{t-1}; \mu_*, \sigma_t^2 \mathbf{I}) \\ \mu_* &= x_t - \alpha_t e - \frac{\beta_t^2}{\bar{\beta}_t} \varepsilon \\ \sigma_t^2 &= \beta_t^2 \bar{\beta}_{t-1}^2 / \bar{\beta}_t^2 \end{aligned} \quad (6)$$

where ε is the white noise added to x_t during the forward process. Equation (6) describes that, in the reverse process, given x_t, e , and ε , we can derive x_{t-1} . Therefore, the reverse process allows for the recovery of x_0 from x_T . We use the

reparameterization method, which leads to further derivation

$$x_{t-1} = x_t - \alpha_t e - \frac{\beta_t^2}{\bar{\beta}_t} \varepsilon + \sigma_t \varepsilon'_t \quad (7)$$

where $\varepsilon'_t \sim \mathcal{N}(0, \mathbf{I})$ represents the Gaussian noise introduced after reparameterization. Through derivation, we modify the reverse process described by (6) into a generation strategy for x_{t-1} given x_t, e , and ε . At the same time, we introduce an ECG-colored noise prediction model $e_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t)$ to replace e in (7), and use a white noise prediction model $\varepsilon_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t)$ to replace ε , thus completing the derivation of the entire reverse process. Here, $\bar{\alpha}_t$ and $\bar{\beta}_t$ provide the positional information of diffusion step t for the prediction models. In this study, to ensure the sufficiency of the predicted information, we consider inferring e and ε given the original ECG input $\tilde{x}$, that is, using the prediction model $e_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x})$ to estimate $p_\theta(e|x_t, t, \tilde{x})$, and using $\varepsilon_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x})$ to estimate $p_\theta(\varepsilon|x_t, t, \tilde{x})$.

C. Training and Denoising Process of EDDM

1) *Optimization Target*: During the reverse process, we predict two noises introduced during diffusion. Therefore, our optimization goal is to ensure the predictions are as accurate as possible. We use L2 loss to measure the differences between them

$$\begin{aligned} L(\theta) &= \mathbb{E} \left[\left\| \varepsilon - \varepsilon_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x}) \right\|^2 \right] \\ &\quad + \mathbb{E} \left[\left\| e - e_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x}) \right\|^2 \right]. \end{aligned} \quad (8)$$

2) *Training Process*: Algorithm 1 presents the training process of the EDDM model. For a batch of training data $\{\tilde{x}, x_0\}$, we first randomly select the diffusion step t and use (5) to sample the signal x_t after t steps of diffusion. Then, we use two noise prediction models to obtain the predicted values of the white and colored noise $\varepsilon_\theta, e_\theta$ added at step t . Subsequently, we calculate the loss function using (8) and optimize the two prediction models based on gradient descent. Algorithm 1 is summarized as follows.

3) *Denoising Process*: The denoising process is implemented based on the reverse process, as shown in Algorithm 2. For a given noisy ECG signal $\tilde{x}$, we assume $x_T = \tilde{x}$, meaning that the clean ECG x_0 is diffused to obtain $\tilde{x}$ after T steps. Subsequently, the reverse process starts from the T th step, calculating x_{T-1} according to (7). Here, in each reverse step, we use two trained models to predict the required e and ε in (7). Algorithm 2 is summarized as follows.

D. Structure of the Noise Prediction Model

In the reverse process of EDDM, it is necessary to iteratively predict the noise signal from x_T to x_0 for T steps, and each step of prediction depends on the previous prediction result. Therefore, when designing the prediction models, we need to avoid excessive information loss and alleviate the accumulation of errors. In this article, we designed the noise prediction models based on two 1-D U-Net. U-Net [37] is a CNN widely used in areas such as image segmentation. Its unique U-shaped structure can down-sample the input signal to the deep feature

Algorithm 1 Training Process of EDDM

Input: noisy ECG and clean ECG $\{\tilde{x}, x_0\}$;
Gaussian noise ε ;

Hyperparameters: time steps T ; coefficient α_T, β_T ;
learning rate γ and training steps M .

Output: the ECG noise model e_θ ,
white noise model ε_θ

1. $e_\theta, \varepsilon_\theta$ and i initialized
2. **while** $i < M$ **do**
3. $t \sim \text{Uniform}(\{1, 2, 3, \dots, T\})$; $\varepsilon \sim N(0, \mathbf{I})$;
 $e = \tilde{x} - x_0$
4. $x_t \sim q(x_t|x_0)$ #Get x_t by (5)
5. $e' = e_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x})$; $\varepsilon' = \varepsilon_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x})$
6. Update using $-\gamma \nabla_\theta L(e', e, \varepsilon', \varepsilon)$ #Get L by (8)
7. $i = i + 1$
8. **End while**
9. **Return** $e_\theta, \varepsilon_\theta$

space, and then restore the compressed features to the original data dimensions. In addition, during the compression and restoration process, skip connections [38] are used to reduce signal information loss during the compression.

As shown in Fig. 2(b), the overall structure of U-Net can be viewed as “encoder–decoder.” In the encoder part, we first use 1-D convolutional layers to map the signal $[x_t, \tilde{x}]$, and use fully connected layers and sinusoidal position embeddings to encode the steps’ position $[\bar{\alpha}_t, \bar{\beta}_t]$. Subsequently, we use four residual “DownBlock” modules to extract features from the input signal and deeply fuse them with step information. By using a “Downsample” layer, the compression of signal features is achieved. It is worth noting that every time the signals undergo time fusion and down-sampling, we use skip-connections to pass them to the corresponding “UpBlock” module, to avoid cumulative information loss. After compression, we employ a middle module based on self-attention and residual convolutional layers to enhance the extraction of temporal information on features. Similarly, the decoder layer is composed of residual convolutional layers and up-sampling layers, where the residual layers merge the corresponding features passed from the encoder layer, and the up-sampling layer consists of 1-D transposed convolution layer to adjust the feature size of the decoded output.

The ECG signal contains characteristic waveforms with varying scales, such as P-QRS-T. The extraction of multi-scale features is challenging in the field of AI-ECG. Recent studies [39], [40], [41] have demonstrated that multiscale convolutional kernels enhance the comprehensive extraction of ECG features. In addition, ECG noises also exhibit significant scale variations. To accurately predict the noise signals, a multiscale information extraction module is required. Therefore, we introduced a DAPP layer [42] at the first skip connection to enhance the expression of information at different scales in the compressed features, as shown in Fig. 3. For the features obtained through the compression module, we use five pooling layers of lengths 3, 5, 9, and global pooling to extract deep features of various granularity from different scales. Then,

Algorithm 2 Denoise Process of EDDM

Input: noisy ECG $\tilde{x}$; the ECG noise model e_θ ;
white noise model ε_θ ; Gaussian noise ε ;

Hyperparameters: time steps T ; coefficient α_T, β_T ;

Output: The denoised ECG $\hat{x}$

1. **Initiate:** $x_T = \tilde{x}, t = T$
2. **while** $t > 0$ **do**
3. $\varepsilon'_t \sim N(0, \mathbf{I})$;
4. $e = e_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x})$; $\varepsilon = \varepsilon_\theta(x_t, \bar{\alpha}_t, \bar{\beta}_t, \tilde{x})$;
5. $x_{t-1} \sim p(x_{t-1}|x_t)$ #Get x_{t-1} by (7)
6. $t = t - 1$
7. **End while**
8. **Return** $\hat{x} = x_0$

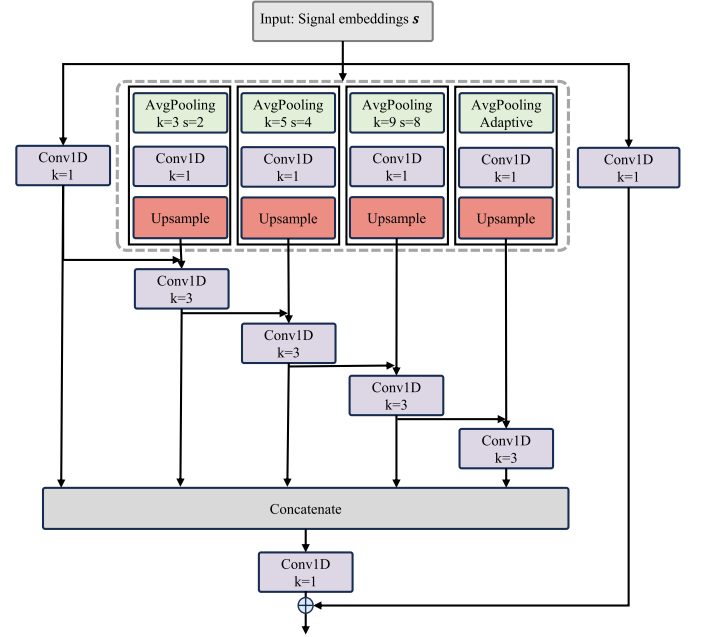


Fig. 3. Structure of DAPP. (k : kernel size.)

we add the adjacent fine-grained and coarse-grained features and use a convolutional layer for feature fusion. To ensure dimensionality consistency before and after fusion, we linearly interpolate the coarse-grained features and add them to the fine-grained features. This approach enhances the capability to extract multiscale features, helping U-Net to better extract target information.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Data Preprocessing

To have a fair comparison of denoising performance, we utilize datasets 1) and 2) consistent with the [25], [33] to train and test EDDM and dataset 3) for external testing.

1) *QT Database*: The QT dataset [43] provides benchmark ECG x_0 for training and testing denoising models. The QT dataset is derived from seven publicly available datasets, containing a total of 105 dual-lead ECG records with a duration of 15 min. These records have a sampling frequency of 250 Hz

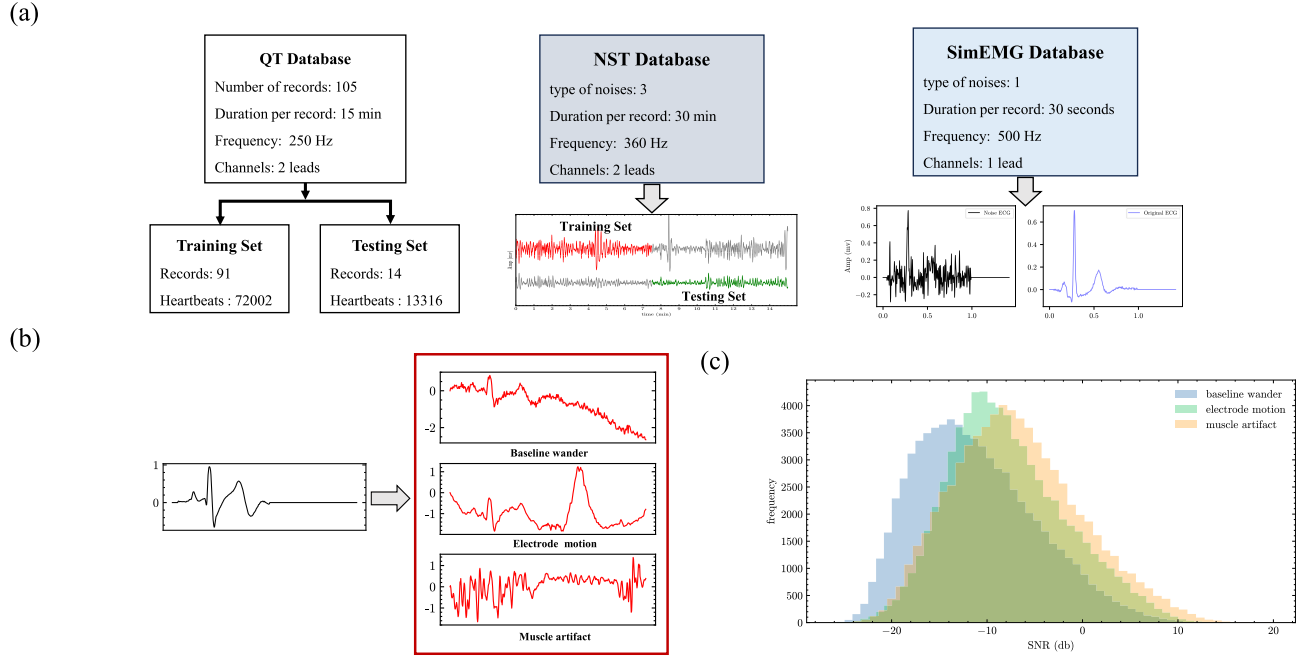


Fig. 4. Datasets splitting and processing. (a) Process of three datasets partitioning. (b) Noisy ECG synthesis. (c) Frequency distribution plot of noisy ECG signals' SNR.

and exhibit no obvious BW or artificial interference noise. The ECG fiducial points are clear. Furthermore, these records encompass various ECG waveforms with different cardiac arrhythmias and ST-T changes, making them highly representative. As a result, the QT database has been widely used in research involving ECG point localization and denoising.

2) *MIT-BIH NST Database:* The noise stress test (NST) database [44] provides three types of ECG noise signals collected from real environments, including BW, MA, and EM. Each noise record consists of two channels with a sampling frequency of 360 Hz and a duration of 30 min. This dataset contains various typical ECG noise waveforms, which can be used to synthesize multiple types of noisy ECG signals $\tilde{x}$ and has been widely utilized in previous ECG denoising research.

3) *SimEMG Database:* The SimEMG dataset [45] contains 110 pairs of ECG recordings collected from real environments. Each pair consists of two signals, each 30 s long, representing ECG with electromyographic (EMG) noise and pure ECG signals, respectively, with a sampling frequency of 500 Hz. According to [46], we select recordings with a noisy ECG signal-to-noise ratio < 5 dB for testing the denoising performance. It is important to emphasize that the ECG and noise domains of this dataset differ from the datasets 1) and 2) used during the training of the model. In this study, we use it to evaluate the cross-dataset capability of the denoising model.

4) *Data Preprocessing:* As shown in Fig. 4(a), we divide the QT dataset and NST into training and testing sets. In the QT dataset, we selected 91 ECG records as the training set, while the remaining 14 records were used as the testing set. For the NST database, the first 15 min of the first channel were assigned as the training set, and the last 15 min of the second channel were assigned as the testing set for each type of noise. During the preprocessing stage, the QT dataset was resampled

to 360 Hz to match the sampling frequency of the NST dataset, ensuring consistency between the datasets. We segmented the ECG records into heartbeats and filled in zero-padding for heartbeats with a length less than 512, resulting in 72 002 heartbeats for the training set and 13 316 heartbeats for the test set. Next, for each ECG segment $x_0 \in R^D$ obtained from the QT dataset, where D represents the signal length, we extracted a corresponding length of noise signal $e \in R^D$ from the training set of NST. We then synthesized the noisy signal $\tilde{x}$ following:

$$\tilde{x} = x_0 + \lambda e$$

$$= \delta \frac{\max(x_0) - \min(x_0)}{\max(e) - \min(e)} \quad (9)$$

where $\delta \in [0.2, 2]$, represents the ratio between the amplitude of the noise e and x_0 . When $\delta = 2$, it indicates that the amplitude range of the noise signal e is twice that of the ECG signal. Fig. 4(b) illustrates the process of synthesizing three types of noisy ECGs from a clean ECG. Unlike the synthesis method based on signal noise ratio (SNR), the strategy that controls the amplitude range yields different SNR distributions for three different noisy ECGs. As shown in Fig. 4(c), we have plotted the SNR distributions of the three noises. Among them, the lowest mean SNR under BW noise is -11.23 dB, the mean SNR for EM noise is -7.88 dB, and the mean SNR for MA is -6.83 dB. For the SimEMG, we adopt the same heartbeat division strategy and this dataset does not use synthetic noise. Ultimately, we obtain 599 pairs of heartbeats for external testing.

B. Evaluation Metrics

To comprehensively evaluate the denoising effect of ECG, we utilized six evaluation metrics commonly used in many

TABLE I
COMPARISON RESULTS IN THE “BW” NOISE (NOISE INTENSITY: $0.2 \leq \delta \leq 2$)

Model	SSD (au) ↓	MAD (au) ↓	PRD (%) ↓	Cosine Sim ↑	SNR out (dB) ↑	ImSNR (dB) ↑
IIR [16]	35.645±70.189	0.594±0.532	61.125±23.220	0.736±0.208	-	-
FCN-DAE [23]	10.633±14.417	0.598±0.381	94.504±70.717	0.770±0.171	-	-
DRNN [32]	6.702±11.316	0.480±0.340	50.035±24.802	0.878±0.126	-	-
Vanilla NL[33]	7.810 ± 11.795	0.431 ± 0.303	63.294±35.386	0.845±0.121	-	-
Multibranch LANLD [33]	5.203±7.915	0.386±0.283	50.445±29.598	0.897±0.097	-	-
DeScoD-1 [25]	4.912±7.822	0.369±0.289	44.218±27.980	0.904±0.113	-	-
DeScoD-3	4.057±6.202	0.340±0.266	41.523±26.169	0.920±0.095	-	-
DeScoD-5	3.887±5.911	0.334±0.261	40.987±26.562	0.923±0.091	-	-
DeScoD-10	3.771±5.713	0.329±0.258	40.527±26.258	0.926±0.087	-	-
EDDM-1	4.522±8.040	0.349±0.257	39.620±20.034	0.919±0.096	9.496±4.167	20.671±6.533
EDDM-3	4.072±7.576	0.328±0.250	37.592±19.592	0.928±0.089	10.090±4.204	21.265±6.549
EDDM-5	3.979±7.381	0.324±0.249	37.170±19.590	0.930±0.087	10.223±4.220	21.398±6.566
EDDM-10	3.920±7.361	0.321±0.248	36.836±19.473	0.932±0.086	10.324±4.235	21.499±6.574

studies [25], [32], [33]. The definitions of these metrics are as follows.

1) *MAD*: This metric measures the Chebyshev Distance between two signals. To avoid the blurring of waveforms in ECG caused by filtering, we use MAD to assess whether the algorithm leads to peak loss. The smaller the MAD, the better the filtered signals

$$\text{MAD} = \max_i (|x_0^i - \hat{x}^i|), \quad i = 1, 2, \dots, D \quad (10)$$

where $\hat{x}$ represents the denoised signal.

2) *PRD*: It calculates the ratio between the Euclidean distance of $\hat{x}$ and x_0 , and the Euclidean distance of $\hat{x}$ and the mean value of x_0 . A smaller value indicates closer similarity between $\hat{x}$ and x_0

$$\text{PRD} = \frac{\sqrt{\sum_{i=1}^D (\hat{x}^i - x_0^i)^2}}{\sqrt{\sum_{i=1}^D (\hat{x}^i - \bar{x}_0)^2}} \times 100\%. \quad (11)$$

3) *Sum of the Square of the Distance (SSD)*: This metric calculates the squared Euclidean distance between $\hat{x}$ and x_0 , and comprehensively represents the differences at all positions of two signals

$$\text{SSD} = \sum_i^D (x_0^i - \hat{x}^i)^2. \quad (12)$$

4) *Cosine Similarity (CosSim)*: This metric is used to calculate the angle between two vectors, with a range of $[-1, 1]$. The closer the value is to 1, the more similar the two signals are

$$\text{CosSim} = \frac{\langle x_0, \hat{x} \rangle}{\|x_0\| \cdot \|\hat{x}\|}. \quad (13)$$

5) *Improved SNR (ImSNR)*: This indicator represents the change in SNR before and after filtering. The higher the SNR improvement achieved by the filtering algorithm, the better the filtering is. In addition, the output SNR (SNR_Out) is $\text{SNR}(x_0, \hat{x})$. These two metrics are defined as follows:

$$\begin{aligned} \text{mSNR} &= \text{SNR}(x_0, \hat{x}) - \text{SNR}(x_0, \bar{x}) \\ \text{SNR}(x_1, x_2) &= 10 \lg \frac{\sum_{i=1}^D (x_1^i)^2}{\sum_{i=1}^D (x_2^i - x_1^i)^2}. \end{aligned} \quad (14)$$

C. Experiments Design and Implementation Details

1) *Experimental Design*: We design three benchmark experiments to evaluate EDDM on three common noises: BW, EM, and MA. For each experiment, we independently generate datasets, divide them into training and testing sets, and train EDDM from scratch before validating it on the test sets. We compare EDDM with six other models, including classical digital filters, DAE-based deep learning models, and conditional diffusion denoising models. The comparative validation results are described in detail in Sections IV-D–IV-F. Meanwhile, in Section IV-G, we present the cross-domain performance comparison of EDDM.

In addition, to thoroughly evaluate EDDM, we provide statistical significance analysis of benchmark experiments results, cross-validation results, and computation time comparisons in Appendixes B–D, respectively.

2) *Implementation Details*: In the EDDM training phase, we generated the ECG noise e diffusion coefficient sequence α_t with a “linear decrease” strategy, and the white noise ε diffusion coefficient sequence with a “linear decrease” strategy [35]. We set the number of diffusion steps to 50, consistent with [25]. The optimizer used is *Radam* with an initial learning rate of $1e^{-5}$, which can adaptively adjust the learning rate during training, and the training batch size is set to 64. The training settings of the comparative models are taken from [33]. Equation (7) denotes that the final denoised signal contains a certain proportion of noise. To more comprehensively compare the denoising effects of the models, we also introduced the multifold ensemble strategy from [25] in our experiments. In the following text, “EDDM- k ” represents the average result obtained by repeating the model’s denoising process on the noisy ECG signal for k time.

D. Result on the BW Noise

The denoising performance of the model on BW noise is presented in Table I. In comparison to the six baseline models, EDDM demonstrates a significantly improved denoising effect ($P < 0.01$). It achieves SOTA results in MAD, PRD, and CosSim. Specifically, EDDM-10 achieves values of 0.321 ± 0.248 (au), 36.836 ± 19.473 (%), and 0.932 ± 0.086 for MAD, PRD, and CosSim, respectively. In addition, EDDM performs

TABLE II
COMPARISON RESULTS IN THE “EM” NOISE (NOISE INTENSITY: $0.2 \leq \delta \leq 2$)

Model	SSD (au) ↓	MAD (au) ↓	PRD (%) ↓	Cosine Sim ↑	SNR out (dB) ↑	ImSNR (dB) ↑
IIR	98.785±143.936	1.012±0.756	79.290±18.746	0.550±0.232	-2.833±5.521	6.2833±4.004
FCN-DAE	14.210±18.222	0.641±0.416	71.958±30.435	0.743±0.209	3.732±3.497	12.849±5.478
DRNN	14.265±15.870	0.785±0.493	136.367±96.619	0.688±0.227	3.235±2.579	12.351±6.072
Vanilla NL	15.520±19.470	0.672±0.436	106.042±62.721	0.692±0.212	3.210±3.040	12.327±5.690
Multibranch LANLD	12.107±14.955	0.600±0.387	98.032±60.615	0.747±0.190	4.242±3.019	13.359±5.533
DeScoD-1	12.415±20.537	0.581±0.436	70.320±44.597	0.768±0.247	4.942±4.132	14.059±6.239
DeScoD-3	10.374±16.739	0.535±0.405	68.796±44.138	0.797±0.219	5.609±3.894	14.726±6.184
DeScoD-5	9.967±15.836	0.525±0.399	68.581±45.214	0.804±0.213	5.769±3.851	14.885±6.200
DeScoD-10	9.650±15.463	0.516±0.395	68.400±46.285	0.810±0.207	5.916±3.818	15.033±6.216
EDDM-1	12.551±21.611	0.542±0.426	60.672±31.008	0.791±0.232	5.281±4.703	14.398±6.072
EDDM-3	11.663±20.733	0.517±0.418	59.181±31.738	0.804±0.225	5.748±4.752	14.865±6.050
EDDM-5	11.454±20.558	0.512±0.416	58.790±31.906	0.807±0.224	5.858±4.763	14.975±6.051
EDDM-10	11.333±20.392	0.508±0.415	58.553±32.072	0.809±0.222	5.936±4.776	15.053±6.056

TABLE III
COMPARISON RESULTS IN THE “MA” NOISE (NOISE INTENSITY: $0.2 \leq \delta \leq 2$)

Model	SSD (au) ↓	MAD (au) ↓	PRD (%) ↓	Cosine Sim ↑	SNR out (dB) ↑	ImSNR (dB) ↑
IIR	77.467±116.400	1.225±0.921	76.063±18.647	0.588±0.210	-1.782±5.419	4.433±4.625
FCN-DAE	8.448±10.852	0.585±0.403	59.055±23.421	0.831±0.130	5.712±2.867	11.927±6.177
DRNN	8.841±11.377	0.629±0.421	76.261±40.904	0.819±0.138	5.432±2.826	11.647±5.971
Vanilla NL	8.010±10.530	0.517±0.383	67.916±38.806	0.831±0.131	6.134±3.262	12.349±5.571
Multibranch LANLD	6.011±7.923	0.489±0.380	57.168±34.036	0.881±0.099	7.385±3.171	13.600±5.703
DeScoD-1	5.470±9.547	0.480±0.378	44.164±23.826	0.900±0.123	8.610±3.986	14.825±5.877
DeScoD-3	4.646±8.287	0.441±0.350	41.586±23.016	0.915±0.108	9.320±0.3920	15.535±5.880
DeScoD-5	4.476±8.028	0.433±0.344	40.961±22.887	0.918±0.104	9.487±3.895	15.702±5.883
DeScoD-10	4.354±7.831	0.422±0.339	40.059±22.809	0.921±0.104	9.613±3.885	15.828±5.889
EDDM-1	4.917±9.665	0.437±0.348	39.452±19.875	0.914±0.111	9.274±4.175	15.488±5.527
EDDM-3	4.465±9.181	0.412±0.340	37.335±19.572	0.923±0.105	9.893±4.242	16.108±5.518
EDDM-5	4.389±9.122	0.408±0.339	36.898±19.590	0.924±0.104	10.029±4.268	16.244±5.512
EDDM-10	4.319±9.028	0.404±0.337	36.525±19.518	0.926±0.103	10.141±4.284	16.356±5.509

equivalently to DeScoD-10 in terms of SSD, while still surpassing nondiffusion-based denoising models. Simultaneously, the multifold ensemble strategy necessitates multiple iterations for processing noisy ECG signals, resulting in significant computational complexity. In the context of real-time ECG monitoring applications, single-fold denoising offers greater advantages and potential for practical applications. Hence, we emphasize the model’s performance specifically under the condition of single-fold denoising. Upon comparison with DeScoD-1 and other nondiffusion denoising models, EDDM-1 has attained notable benefits and exhibited superior performance across all evaluation metrics. EDDM-1, compared to DeScoD-1, improves by 7.94% in SSD ($P < 0.01$). It is equivalent to DeScoD-3 in MAD and CosSim and even surpasses DeScoD-10 in PRD. This result suggests that, due to the dual-path noise diffusion process, EDDM establishes a diffusion path from the clean ECG to the BW noise ECG, thus achieving better denoising performance.

E. Result on the EM Noise

Table II shows the denoising performance and comparative results of the model on EM noise. EDDM-10 still maintains excellent denoising performance, achieving SOTA results ($P < 0.01$) of 0.508 ± 0.415 (au) for MAD, 58.553 ± 32.072 (au) for PRD, 5.936 ± 4.776 (dB) for SNR_Out, and

15.053 ± 6.056 (dB) for ImSNR. It is worth mentioning that EDDM-10 significantly improves PRD performance by 9.85% compared to DeScoD-10 ($P < 0.01$).

Regarding the single-fold performance, EDDM-1 demonstrates superiority over DeScoD-1 when it comes to EM noise. The model closely aligns with DeScoD-3 in MAD and CosSim, while surpassing DeScoD-10 by 7.73% in the PRD index. In a word, the experimental results show the exceptional denoising capabilities of EDDM in the presence of EM noise. It outperforms the current benchmark models and achieves SOTA results in four metrics.

F. Result on the MA Noise

In Table III, we present the denoising results of the MA noise. EDDM outperforms six benchmark models in both single-fold and multiple-fold denoising. Specifically, EDDM-10 achieves SOTA results in all metrics, with SSD, MAD, PRD, CosSim, and ImSNR reaching 4.319 ± 9.028 (au), 0.404 ± 0.337 (au), 36.525 ± 19.518 (%), 0.926 ± 0.103 , and 16.356 ± 5.509 (dB), respectively. In the single-fold denoising, EDDM-1 also consistently outperforms DeScoD-1 and the other five denoising models, indicating its excellent performance ($P < 0.01$). This experiment further demonstrates that EDDM remains effective and performs strongly against MA noise.

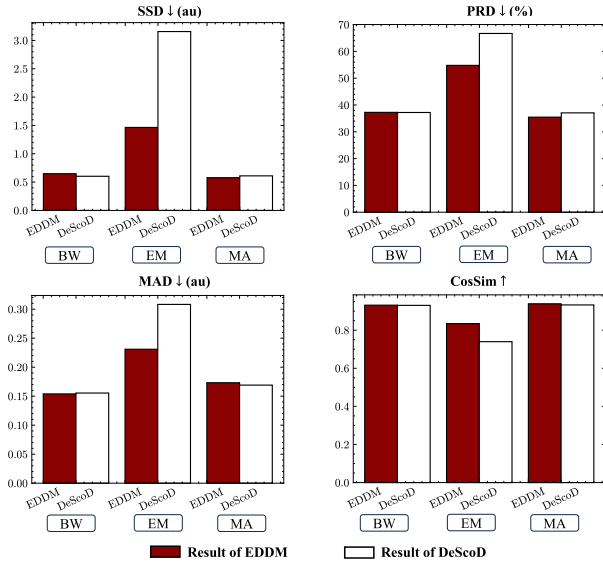


Fig. 5. Comparison of cross-dataset diagnostic performance between EDDM and DeScoD on the SimEMG. “BW” indicates that the model trained on the BW noise set is directly tested for denoising on the SimEMG, with the same logic applied for “EM” and “MA”.

G. Results on the External Dataset

Fig. 5 presents a comparison of the onefold test results of the EDDM model and DeScoD on the SimEMG dataset. The EDDM model exhibits superior performance across four denoising metrics on the SimEMG dataset, maintaining resilience against variations in both the ECG and noise domains. In comparison with DeScoD, the EDDM model performs as well as or better in almost all evaluations across three cross-domain experiments and four metrics. Notably, in the EM cross-domain experiment, EDDM achieved marked improvements over DeScoD. We attribute this to EDDM’s distinctive dual-path mechanism, which facilitates a more comprehensive separation of noise and signal during training, effectively recognizing the ECG. Fig. 4(a) and (b) demonstrates that, unlike BW and MA, EM noise is characterized by random and intense jitter, distinguishing its noise pattern significantly from EMG noise. The high-frequency spike pattern of EMG is not distinctly evident in EM noise. We conclude that during the reverse process, the DeScoD model predicts diffusion noise $\varepsilon_\theta(x_t, \tilde{x})$ using noisy ECG $\tilde{x}$ (conditioned on EMG noise, which substantially differs from the original training domain) to derive x_{t-1} , resulting in substantial errors. In contrast, the EDDM’s dual-path mechanism focuses on isolating ECG signals from noisy ECG signals while predicting ECG noise $e_\theta(x_t, \tilde{x})$, thereby extracting more ECG information and noise components e , which reduces cumulative errors during the reverse process, leading to a notable enhancement over DeScoD.

V. DISCUSSION

A. Dual-Path Diffusion Process Achieves a Transition From Noisy ECG to Clean ECG

The modeling principles of EDDM and DeScoD in ECG denoising are different. EDDM diffuses the signal from the

ECG domain to the real noise domain, while DeScoD spreads the signal to the theoretical noise domain. Fig. 6(a) and (b), respectively, shows the changes of the output signals every ten steps in denoising with EDDM and DeScoD to fully demonstrate their differences. EDDM gradually removes a certain proportion of BW noise and white noise from the noise signal $\tilde{x}$, while retaining the clean ECG. Each step’s result can be considered as obtaining a real ECG signal containing a certain proportion of noise. In comparison to the DeScoD shown in Fig. 6(b), which progressively removes noise from Gaussian white noise to obtain an ECG signal, the result of each step does not belong to the domain of ECG signals. Therefore, EDDM, based on the diffusion of clean ECG into noisy ECG, ensures that each step’s result in the reverse path is interpretable, progressively denoising from highly noisy ECG to clean ECG.

Furthermore, in Fig. 6(c), we respectively plot the changes in the signal’s SNR of EDDM and DeScoD with each denoising step under three types of noise. It can be observed that in the denoising task under three types of noise, the initial signal SNR of EDDM is -11.23 , -7.88 , and -6.83 dB, corresponding to the mean SNR values in Fig. 6(c). As the denoising proceeds step by step, the SNR of the output signal gradually increases until obtaining a clean ECG in the 50th step, which corresponds to the SNR_Out value of the final denoised ECG in Tables I–III. For DeScoD, all its initial signals are -15 dB. This indicates that in different denoising tasks, the initial data domain of DeScoD is in the Gaussian domain. Under the guidance of the given noisy signal $\tilde{x}$, possible ECG signals are gradually generated from Gaussian white noise.

Therefore, we believe that with the modeling approach of the dual-path noise diffusion process, EDDM achieves ECG denoising from the ECG noise domain to the clean ECG domain, which corresponds to the problem definition addressed in Section III-A.

B. DAPP Improves Accuracy of EDDM in Identifying the Amplitude and Position of ECG Waveforms

From Fig. 7(a) and (b), it can be seen that the DAPP module improves the model’s performance on SSD while maintaining excellent performance on other metrics. In the ten-fold ensemble, the model using the DAPP module improved the SSD metric by more than 5.3%. In the single-fold denoising, EDDM achieves a reduction in the SSD value from 4.756 ± 9.092 (au) to 4.522 ± 8.040 (au), corresponding to a decrease of over 4.9%. The SSD metric comprehensively evaluates the discrepancy between the denoised signal and the true signal at each position. Therefore, we speculate that multiscale information helps the model observe the entire input signal from multiple scales, capture the noise trend of the noisy ECG, and thereby more accurately estimate the signal trend of the ECG, thus improving the SSD.

As shown in Fig. 8(a) and (b), due to the presence of BW noise, the S-wave to T-wave region in the noisy ECG is either elevated or depressed overall by the noise. Therefore, to restore the ECG, it is necessary to consider the positions of the P-Q-R waves on a global scale and thus estimate the true

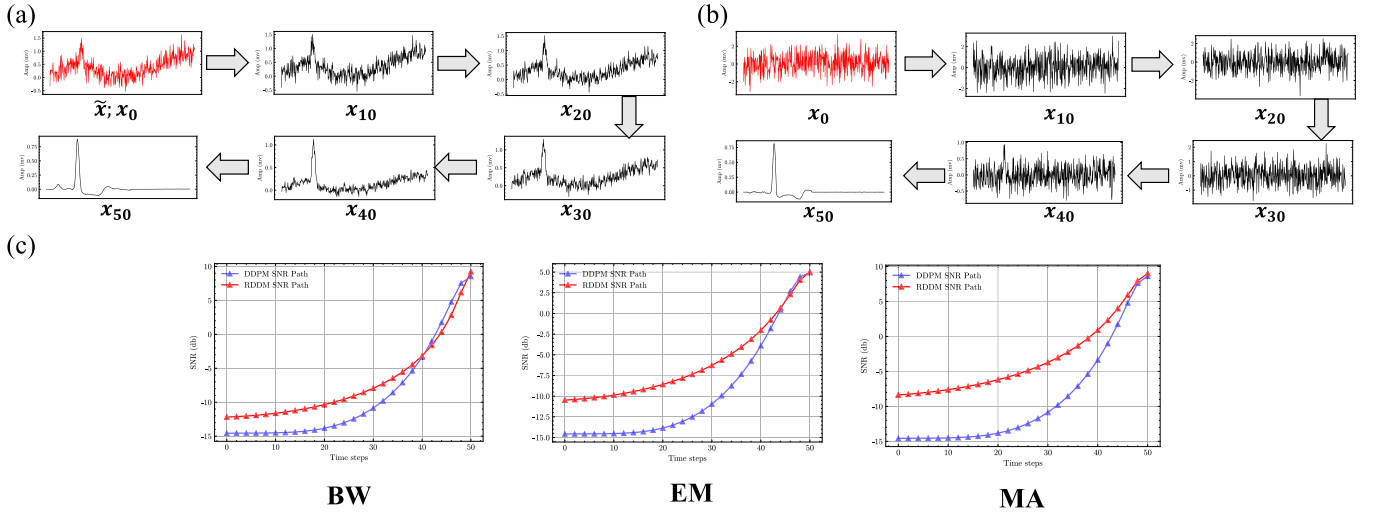


Fig. 6. Visualization of the outputs of EDDM and DeScoD during the reverse process. (a) Output signals of EDDM every ten steps. (b) Output signals of DeScoD every ten steps. (c) Visualization of SNR during the reverse process.

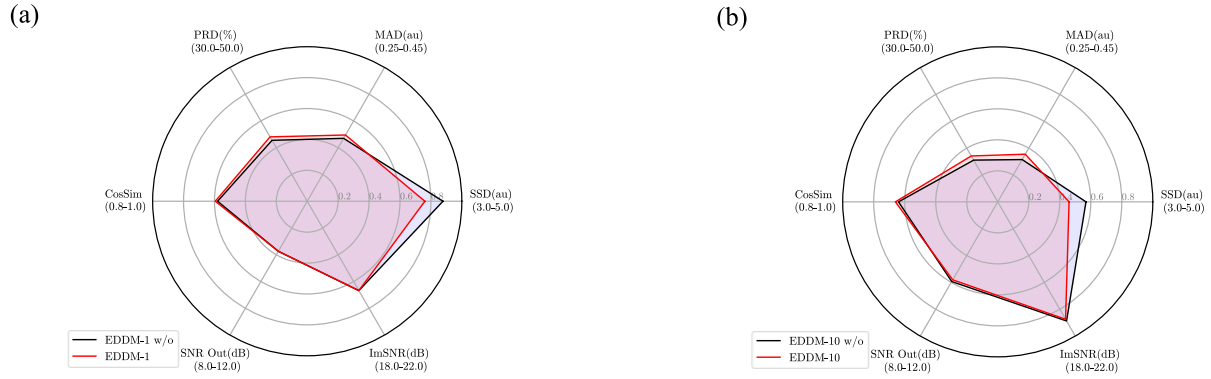


Fig. 7. Radar plots of DAPP Ablation Experiments. “W/o” indicates that the model does not use DAPP. (a) EDDM-1. (b) EDDM-10.

amplitude and position of the T-wave. Multiscale information can better assist EDDM in locating this information, so in the denoised signal obtained, the T-wave amplitude is closer to the true value. For Fig. 8(c) and (d), the T-wave amplitude requires estimation of the noise in the ST-T region. In addition, since there are many subtle variations of noise on the T-wave, small-scale information can play a positive role, making the T-wave obtained by EDDM more accurate. Therefore, multiscale information can help the EDDM model achieve position localization and amplitude estimation of ECG characteristic waveforms in various noises, thus significantly improving the SSD metric.

C. EDDM Maintains Excellent Performance Under Different Noise Intensities

Tables IV–VII show the comparison results of four metrics between EDDM and six benchmark models under different noise intensities. Overall, the EDDM model significantly outperforms all benchmark models, including DeScoD, in all metrics under nonextreme noise conditions (0.2–1.0), achieving superior results. EDDM removes noise directly from the noisy ECG, so when it is easier to perceive ECG information

in the noisy ECG, EDDM can accurately restore the ECG signal. In contrast, DeScoD’s denoising process is essentially a conditional generative model, where the denoising process removes components from a Gaussian white noise and generates the ECG signal. Therefore, in the noise range of 0.2–1.0, the DeScoD model does not directly use the obvious ECG information in the noisy signal but generates the ECG based on a conditional generation strategy, making it less effective than EDDM. Similarly, under extremely high noise conditions, where the ECG information is almost completely submerged in noise, the generation strategy can produce signals that more closely resemble the ECG. EDDM, however, performs noise and ECG separation based on extreme noise, using relatively little ECG information, which leaves some noise in the denoised signal. Thus, we speculate that this is the reason why there is a certain gap in the SSD metric between EDDM-10 and DeScoD-10, their CosSim metrics are very close.

Comparing single-fold performance, EDDM-1 outperforms DeScoD-1 and the other five benchmark models under different noise scales. Under extremely high noise conditions, except for the 1.5–2.0 range where the SSD metric is close to DeScoD-1, it is far superior in MAD, PRD, and

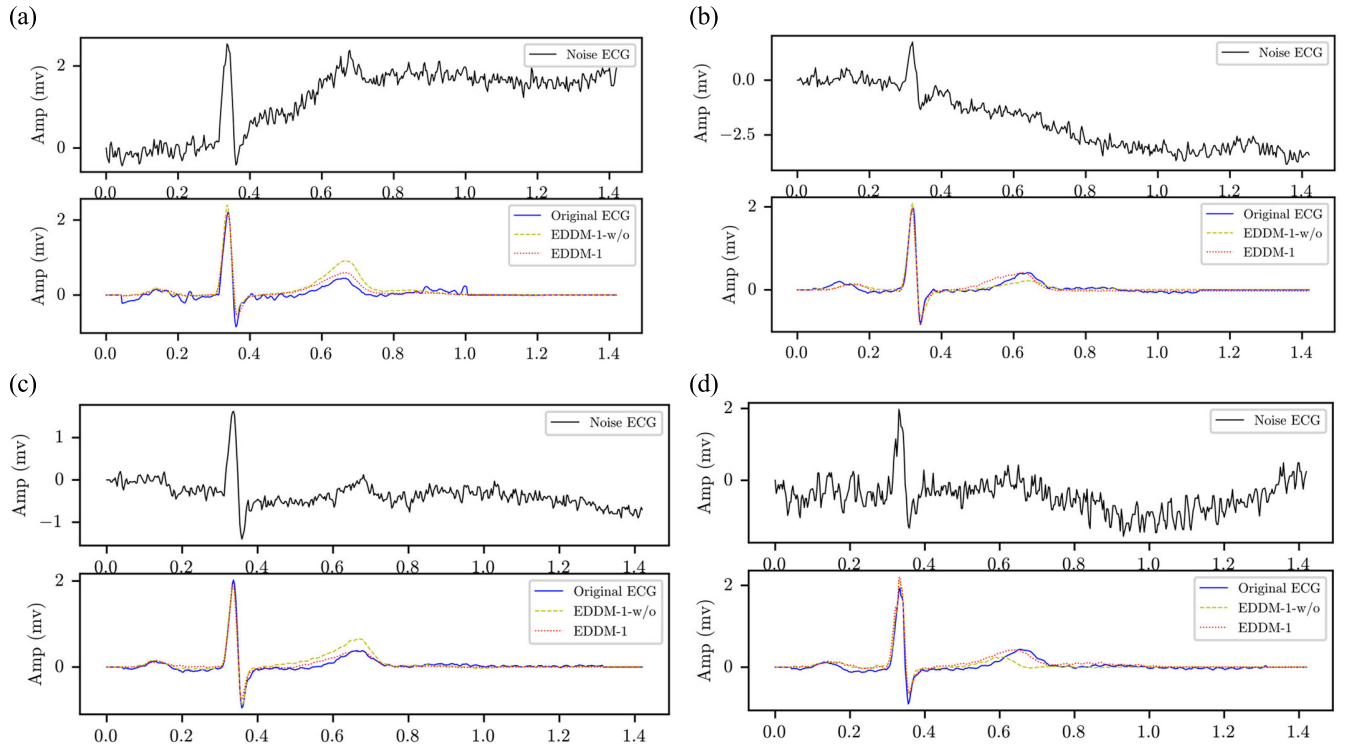


Fig. 8. Visualization of four case studies comparing the denoising performance of EDDM-1-w/o and EDDM-1. (a) Upward fluctuating BW noise. (b) Downward fluctuating BW noise. (c) Low amplitude MA noise. (d) High amplitude MA noise.

TABLE IV
SSD COMPARISON RESULTS UNDER DIFFERENT INTENSITIES OF BW NOISE

Model	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
IIR	5.615±7.508	17.262±25.049	38.334±58.050	71.980±107.880
FCN-DAE	7.396±8.032	8.638±8.247	10.929±12.883	14.586±21.429
DRNN	4.000±7.437	4.932±6.652	7.016±10.971	10.031±15.665
Vanilla NL	4.356±7.463	6.140±7.126	8.489±11.492	11.267±16.273
M_LANLD	2.401±3.419	3.816±4.771	5.717±7.781	8.078±11.104
DeScoD-1	2.324±3.798	3.762±5.869	5.328±7.195	7.466±10.724
DeScoD-3	2.036±3.225	3.252±5.261	4.421±5.739	5.935±8.164
DeScoD-5	1.977±3.201	3.147±4.963	4.193±5.474	5.686±7.772
DeScoD-10	1.953±3.138	3.046±4.902	4.081±5.315	5.483±7.433
EDDM-1	1.810±3.113	3.041±3.945	4.752±7.127	7.795±13.114
EDDM-3	1.612±2.871	2.716±3.660	4.279±6.475	6.987±11.923
EDDM-5	1.577±2.834	2.648±3.571	4.221±6.397	6.800±11.522
EDDM-10	1.538±2.786	2.603±3.574	4.135±6.313	6.702±11.475

TABLE V
MAD COMPARISON RESULTS UNDER DIFFERENT INTENSITIES OF BW NOISE

Model	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
IIR	0.248±0.171	0.449±0.322	0.666±0.487	0.918±0.676
FCN-DAE	0.480±0.278	0.532±0.301	0.617±0.369	0.727±0.474
DRNN	0.390±0.286	0.432±0.273	0.496±0.330	0.576±0.409
Vanilla NL	0.319±0.220	0.377±0.224	0.453±0.289	0.544±0.381
M_LANLD	0.293±0.214	0.341±0.222	0.401±0.270	0.483±0.351
DeScoD-1	0.230±0.188	0.321±0.225	0.403±0.275	0.484±0.351
DeScoD-3	0.216±0.180	0.299±0.213	0.369±0.251	0.442±0.323
DeScoD-5	0.213±0.178	0.295±0.211	0.362±0.247	0.432±0.314
DeScoD-10	0.211±0.178	0.291±0.210	0.356±0.243	0.426±0.313
EDDM-1	0.215±0.161	0.295±0.197	0.378±0.244	0.471±0.314
EDDM-3	0.201±0.154	0.277±0.193	0.357±0.239	0.442±0.304
EDDM-5	0.198±0.152	0.274±0.192	0.353±0.238	0.436±0.301
EDDM-10	0.196±0.151	0.271±0.192	0.349±0.237	0.432±0.302

CosSim, with a more than 6% improvement in PRD compared to DeScoD-1. Fig. 9(a) and (b) shows the performance of EDDM and DeScoD under the multifold ensemble strategy. Extreme high noise submerges crucial ECG information such

TABLE VI
PRD COMPARISON RESULTS UNDER DIFFERENT INTENSITIES OF BW NOISE

Model	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
IIR	38.653±15.668	55.867±19.482	67.972±19.714	76.644±18.290
FCN-DAE	76.445±41.696	86.350±56.163	98.331±68.010	112.012±94.656
DRNN	40.298±19.983	45.582±21.936	52.317±24.336	59.312±27.389
Vanilla NL	46.274±25.272	56.990±29.432	67.685±34.481	77.698±40.459
M_LANLD	34.026±14.234	43.346±19.925	53.777±27.189	66.167±38.040
DeScoD-1	29.762±15.583	38.945±20.183	47.678±25.366	56.559±35.969
DeScoD-3	28.066±14.923	36.681±19.165	44.506±23.090	53.225±33.843
DeScoD-5	27.697±14.706	36.187±18.974	43.774±22.834	52.715±35.341
DeScoD-10	27.475±14.631	35.710±18.718	43.321±22.959	52.081±34.681
EDDM-1	26.425±12.655	35.364±15.271	42.893±19.024	50.251±22.752
EDDM-3	24.660±11.735	33.435±14.601	40.782±18.535	47.957±22.541
EDDM-5	24.337±11.645	33.006±14.553	40.436±18.594	47.494±22.568
EDDM-10	24.035±11.440	32.670±14.622	39.990±18.453	47.117±22.479

TABLE VII
COSIM COMPARISON RESULTS UNDER DIFFERENT INTENSITIES OF BW NOISE

Model	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
IIR	0.910±0.074	0.800±0.143	0.690±0.191	0.589±0.217
FCN-DAE	0.844±0.110	0.807±0.136	0.757±0.167	0.692±0.205
DRNN	0.926±0.070	0.906±0.084	0.871±0.122	0.824±0.167
Vanilla NL	0.916±0.062	0.875±0.086	0.829±0.115	0.778±0.146
M_LANLD	0.948±0.042	0.921±0.064	0.888±0.093	0.844±0.124
DeScoD-1	0.955±0.049	0.929±0.068	0.895±0.104	0.853±0.156
DeScoD-3	0.961±0.042	0.939±0.058	0.913±0.083	0.879±0.132
DeScoD-5	0.961±0.041	0.941±0.056	0.918±0.078	0.883±0.129
DeScoD-10	0.962±0.040	0.943±0.055	0.920±0.077	0.889±0.124
EDDM-1	0.967±0.034	0.942±0.054	0.913±0.086	0.868±0.136
EDDM-3	0.972±0.028	0.950±0.046	0.922±0.079	0.882±0.127
EDDM-5	0.973±0.028	0.951±0.045	0.924±0.079	0.885±0.125
EDDM-10	0.974±0.027	0.952±0.045	0.926±0.078	0.886±0.124

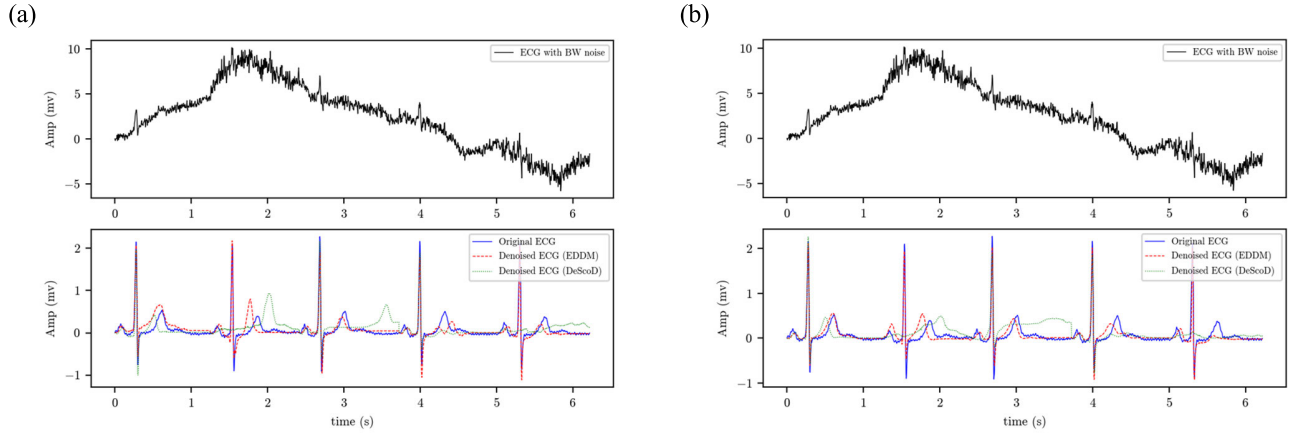


Fig. 9. Visualization of the denoising results between EDDM and DeScoD under extreme noise. (a) Single-fold. (b) Ten-fold.

as P-waves and T-waves. The DeScoD generates positions for these waveforms, as seen in the two T-waves between 2 and 4 s in Fig. 9(a). Therefore, the single-fold estimation is highly uncertain and performs worse than EDDM. In the ten-fold denoising process, the output results of DeScoD undergo significant changes compared to the single-fold, such as the T-wave at 2 s becoming closer to the true value and reduced fluctuation at 6 s. Therefore, we note that under extremely high noise conditions, EDDM is more robust to the multifold integration strategy, meaning that the proportion of random noise in the denoised results of EDDM is lower, and the output is more stable.

VI. CONCLUSION AND FUTURE WORKS

In this work, we propose a novel EDDM, named EDDM. In EDDM, we establish a directional dual-path diffusion process capable of concurrently diffusing both ECG noise and Gaussian white noise. This process facilitates the transformation between clean and noisy ECG. Compared to classical generative diffusion models, EDDM can achieve a stable and interpretable denoising process. In addition, we design a 1-D residual U-Net incorporating a deep aggregated pyramid pooling module, which identifies ECGs and predicts noise from a multiscale perspective, thereby enhancing denoising performance. Compared with six benchmark

ECG denoising models, EDDM achieved the best results on five metrics across three benchmark experiments, and it demonstrates stable cross-domain denoising performance. Therefore, EDDM will become the latest benchmark model in the field of ECG denoising. Moreover, EDDM exhibits superior and stable results in extreme noise denoising and has great potential for application in wearable ECG measurement systems.

EDDM still has two main limitations: 1) during the reverse process, EDDM requires multistep noise predictions to generate intermediate signals, thereby ensuring that the noisy ECG is transformed into clean ECG along an interpretable path; and 2) the performance of EDDM on various mixed noises needs further exploration, limited by the scale of existing datasets and the difficulty of measurement, the performance of end-to-end denoising models in open scenarios needs further verification. In the future, we will continue to promote the practical application of EDDM in ECG denoising from the following two aspects. First, we will improve the denoising efficiency of EDDM. Recent studies [47], [48] have shown that the diffusion model can still demonstrate excellent performance under non-Markov Chain conditions. Therefore, we will investigate non-Markovian-based EDDM to improve its reverse sampling efficiency and denoising speed. Second, we will collect and test a wider range of ECG noise datasets based on the wearable ECG measurement system using the Internet of Things (IoT) technology [8], [49], to improve the denoising effect of EDDM. Currently, our team has developed an out-of-hospital 24/7 dynamic ECG monitoring system [50], which can be used for data collection in multiple scenarios, providing a foundation for EDDM in ECG denoising range and performance improvement.

APPENDIX

A. Derivation in the Reverse Process of EDDM

In this section, we have provided a detailed derivation process for (6), as shown below.

According to (5), we provide the relationship in the forward diffusion process, which is given by the following equation:

$$\begin{aligned} q(x_t|x_{t-1}, x_0, e) &= \mathcal{N}(x_t; x_{t-1} + \alpha_t e, \beta_t \mathbf{I}) \\ q(x_t|x_0, e) &= \mathcal{N}(x_t; x_0 + \bar{\alpha}_t e, \bar{\beta}_t \mathbf{I}). \end{aligned} \quad (15)$$

According to Bayes' theorem, we have

$$p(x_{t-1}|x_t, x_0, e) = q(x_t|x_{t-1}, x_0, e) \frac{q(x_{t-1}|x_0, e)}{q(x_t|x_0, e)}. \quad (16)$$

In (16), the reverse relationship from x_t to x_{t-1} is transformed into a combination of three forward relationships, assuming it follows a normal distribution:

$$p(x_{t-1}|x_t, x_0, e) = \mathcal{N}(x_{t-1}; \mu_*, \sigma_t^2 \mathbf{I}). \quad (17)$$

To calculate the reverse process, we need to know the mean μ_* and variance σ_t^2 of this distribution for sampling.

TABLE VIII
T-TEST RESULTS OF THE "BW" EXPERIMENT

Models	1 fold	3 folds	5 folds	10 folds
SSD (au)	P<0.01	P<0.01	P<0.01	P<0.01
MAD (au)	P<0.01	P<0.01	P<0.01	P<0.01
PRD (%)	P<0.01	P<0.01	P<0.01	P<0.01
COS_SIM	P<0.01	P<0.01	P<0.01	P<0.01
SNR_out (dB)	-	-	-	-
ImSNR (dB)	-	-	-	-

Since all three terms in (16) follow a normal distribution, we have:

$$\begin{aligned} p(x_{t-1}|x_t, x_0, e) &\propto \exp\left(-\frac{1}{2}\left[\frac{(x_{t-1} - x_0 - \bar{\alpha}_{t-1}e)^2}{\bar{\beta}_{t-1}^2} + \frac{(x_t - x_{t-1} - \alpha_t e)^2}{\beta_t^2} - \frac{(x_t - x_0 - \bar{\alpha}_t e)^2}{\bar{\beta}_t^2}\right]\right) \\ &= \exp\left(-\frac{1}{2}\left[\left(\frac{1}{\bar{\beta}_{t-1}^2} + \frac{1}{\beta_t^2}\right)x_{t-1}^2 - 2\left(\frac{x_0 + \bar{\alpha}_{t-1}e}{\bar{\beta}_{t-1}^2} + \frac{x_t - \alpha_t e}{\beta_t^2}\right) \times x_{t-1} + f(x_t, x_0, e)\right]\right). \end{aligned} \quad (18)$$

In (18), since the term $f(x_t, x_0, e)$ is unrelated to x_{t-1} , it is also unrelated to the μ_* and σ_t^2 that need to be derived. As for σ_t^2 , since $(1/(\bar{\beta}_{t-1}^2)) + (1/\beta_t^2) = ((\bar{\beta}_{t-1}^2 + \beta_t^2)/(\beta_t^2 \bar{\beta}_{t-1}^2)) = (\bar{\beta}_t^2/(\beta_t^2 \bar{\beta}_{t-1}^2))$, we have

$$\sigma_t^2 = \beta_t^2 \bar{\beta}_{t-1}^2 / \bar{\beta}_t^2. \quad (19)$$

Besides, μ_* can be derived from the coefficient of x_{t-1}

$$\begin{aligned} \mu_* &= \left(\frac{x_0 + \bar{\alpha}_{t-1}e}{\bar{\beta}_{t-1}^2} + \frac{x_t - \alpha_t e}{\beta_t^2}\right) \frac{\beta_t^2 \bar{\beta}_{t-1}^2}{\bar{\beta}_t^2} \\ &= \frac{\beta_t^2}{\bar{\beta}_t^2} (x_0 + \bar{\alpha}_{t-1}e) + \frac{\bar{\beta}_{t-1}^2}{\bar{\beta}_t^2} (x_t - \alpha_t e) \\ &= \frac{\beta_t^2}{\bar{\beta}_t^2} (x_t - \bar{\alpha}_t e - \bar{\beta}_t \varepsilon + \bar{\alpha}_{t-1}e) + \frac{\bar{\beta}_{t-1}^2}{\bar{\beta}_t^2} (x_t - \alpha_t e) \\ &= \frac{\beta_t^2 + \bar{\beta}_{t-1}^2}{\bar{\beta}_t^2} x_t - \frac{\beta_t^2 + \bar{\beta}_{t-1}^2}{\bar{\beta}_t^2} \alpha_t e - \frac{\beta_t^2}{\bar{\beta}_t^2} \varepsilon \\ &= x_t - \alpha_t e - \frac{\beta_t^2}{\bar{\beta}_t^2} \varepsilon. \end{aligned} \quad (20)$$

B. Statistical Significance Analysis of Comparison Results in Three Benchmark Experiments

In this section, we use the T-test to evaluate the comparative results of EDDM and DeScoD in three benchmark experiments.

For a fair comparison, we adopt a testing strategy of "DeScoD- k versus RDDM- k ." As shown in Tables VIII-X, at a significance level of 0.01, in the BW experiment, the

TABLE IX
T-TEST RESULTS OF THE “EM” EXPERIMENT

Models Metrics	1 fold	3 folds	5 folds	10 folds
SSD (au)	P=0.77	P<0.01	P<0.01	P<0.01
MAD (au)	P<0.01	P<0.01	P<0.01	P<0.01
PRD (%)	P<0.01	P<0.01	P<0.01	P<0.01
COS_SIM	P<0.01	P<0.01	P=0.18	P=0.56
SNR_out (dB)	P<0.01	P<0.01	P<0.01	P=0.34
ImSNR (dB)	P<0.01	P<0.01	P<0.01	P=0.34

TABLE X
T-TEST RESULTS OF THE “MA” EXPERIMENT

Models Metrics	1 fold	3 folds	5 folds	10 folds
SSD (au)	P<0.01	P=0.03	P=0.31	P=0.88
MAD (au)	P<0.01	P<0.01	P<0.01	P<0.01
PRD (%)	P<0.01	P<0.01	P<0.01	P<0.01
COS_SIM	P<0.01	P<0.01	P<0.01	P<0.01
SNR_out (dB)	P<0.01	P<0.01	P<0.01	P<0.01
ImSNR (dB)	P<0.01	P<0.01	P<0.01	P<0.01

TABLE XI
COMPARISON OF THE TWO RESULTS ON THE BW NOISE

Metrics	Exp	Original results	10-folds results
SSD (au) ↓		4.522	6.402±2.274
MAD (au) ↓		0.349	0.345±0.064
PRD (%) ↓		39.620	41.020±3.543
Cosine Sim ↑		0.919	0.913±0.012
SNR out (dB) ↑		9.496	9.498±0.499
ImSNR (dB) ↑		20.671	20.454±0.908

differences in results between EDDM and DeScoD are statistically significant across four metrics. In the EM experiment, the P-value for the single-fold denoising SSD results was 0.77, indicating no statistical significance between them, and in the fivefold model, the P-values for CosSim from the fivefold, and CosSim, SNR_Out, and ImSNR from ten-fold are both greater than 0.01, indicating no statistical significance. In the MA experiment, the SSD metric showed no statistical significance in the results across multiple folds.

C. Cross-Validation Results of EDDM

We utilize the partition method described in [25] and [33] to design benchmark experiments for a fair comparison of the denoising effects of EDDM and existing models. In this section, we employ ten-fold cross-validation on the BW dataset to calculate the average denoising performance of EDDM, thereby verifying the rationality of the data partitioning method used in the study.

As shown in Table XI: The average results of the ten-fold cross-validation on the six indicators are comparable to the results of the data division used in the manuscript, where both SSD and PRD are within one standard deviation, and the other four indicators are almost identical. Therefore, we believe that the data division used in this article can reasonably comprehensively evaluate the denoising results of EDDM, and the ten-fold cross-validation demonstrates the stable denoising performance of EDDM.

D. Comparison of Inference Time of Models

We evaluate the inference time consumption of EDDM and comparative models on standard equipment (Intel¹ Xeon¹ Platinum 8168 CPU at 2.70 GHz, NVIDIA GeForce RTX 3090). The inference results show that for a 10-s ECG signal, the average processing time of EDDM is 0.16 s, which can fully meet the real-time requirements of intelligent ECG monitoring. At the same time, for DeScoD, which belongs to the diffusion model, its DeScoD-3 model is comparable to EDDM in denoising performance, while the average time taken to denoise a 10-s ECG is 0.26 s. Therefore, EDDM still has an advantage in terms of model computational time compared to the existing SOTA model.

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